

30-Minute Skill Assessment Report

Virtual Environments, NumPy & Git Workflow

Student Roll No:	pu02324eugbtcs097
Project Directory:	numpy_eval_pu02324eugbtcs097
Deliverable File:	matrix_ops.ipynb (Executed & Merged)
Assessment Status:	100% Completed & Verified

Part 1: Environment, Git Setup & Branching

- 1. Local Directory & Git Repo:** Project folder numpy_eval_pu02324eugbtcs097 was initialized with local Git version control.
- 2. Isolated Virtual Environment:** Python virtual environment venv was created and activated inside the root directory. Installed numpy, jupyter, ipykernel, and nbformat.
- 3. Git Ignore Configuration:** Configured .gitignore to exclude virtual environments, Python cache files, and Jupyter checkpoints.
- 4. Feature Branching:** Created and switched to feature branch feature/matrix-multiply.

```
# Virtual environment directory
venv/
.venv/

# Python bytecode cache
__pycache__/
*.pyc

# Jupyter checkpoint folders
.ipynb_checkpoints/
```

Part 2: NumPy Matrix Dot Product in Jupyter (matrix_ops.ipynb)

Task A: Seed & Matrix Generation

Code Cell:	<pre>import numpy as np np.random.seed(42) # Generate Matrix A (4, 3) and Matrix B (3, 5) with integers in [1, 10] A = np.random.randint(1, 11, size=(4, 3)) B = np.random.randint(1, 11, size=(3, 5)) print("Matrix A (4, 3):\n", A, "\nShape:", A.shape) print("Matrix B (3, 5):\n", B, "\nShape:", B.shape)</pre>
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Executed Output:	Matrix A: <pre>[[7 4 8] [5 7 10] [3 7 8] [5 4 8]]</pre> Shape of Matrix A: (4, 3) Matrix B: <pre>[[8 3 6 5 2] [8 6 2 5 1] [10 6 9 1 10]]</pre> Shape of Matrix B: (3, 5)
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Task B: Dimension Validation & Dot Product

Code Cell:	<pre># Validate inner dimensions assert A.shape[1] == B.shape[0], "Dimension mismatch!" # Compute Matrix Dot Product C = A . B C = np.dot(A, B) print("Resulting Matrix C (A . B):\n", C) print("Shape of Matrix C:", C.shape) assert C.shape == (4, 5)</pre>
Executed Output:	Assertion Passed: Inner dimensions match (A.shape[1] == B.shape[0])! Resulting Matrix C (A . B): <pre>[[168 93 122 63 98] [196 117 134 70 117] [160 99 104 58 93] [152 87 110 53 94]]</pre> Shape of Matrix C: (4, 5) Assertion Passed: Matrix C shape verified as (4, 5).

Task C: Reductions & Summary Statistics

Code Cell:	<pre># Total sum of all elements in C (ΣΣ C_ij) total_sum = np.sum(C) # Row-wise sums (axis=1) row_sums = np.sum(C, axis=1) # Column-wise sums (axis=0) col_sums = np.sum(C, axis=0)</pre>
Executed Output:	Total Sum of all elements in Matrix C (ΣΣ C_ij): 2188 Row-wise sums (axis=1): [544 634 514 496] Column-wise sums (axis=0): [676 396 470 244 402]

Part 3: Git Commit, Branching & Merge History

- 1. Commit on Feature Branch:** Staged .gitignore and matrix_ops.ipynb on feature/matrix-multiply with exact message: feat: implement random matrix creation and dot multiplication
- 2. Merge into Main:** Switched back to main branch and performed a clean non-fast-forward merge.

```
* cced56c Merge branch 'feature/matrix-multiply' into main (main)
| \
| * 5c8a6ba feat: implement random matrix creation and dot multiplication (feature/matrix-multiply)
| /
* 272b37f Initial commit: project structure and gitignore
```

Part 4: Remote Setup & Push Instructions

To publish your completed assessment repository to GitHub or GitLab:

```
git remote add origin https://github.com/<your-username>/numpy-eval-pu02324eugbtcs097.git
git push -u origin main
git push origin feature/matrix-multiply
```